MANIANANTH M

Software Engineer — Python Backend & Full-Stack Architect

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PROFESSIONAL SUMMARY

Production-proven Software Engineer with ~3 years of commercial experience specializing in high-concurrency Python & FastAPI backends, asynchronous microservice architectures (AsyncIO), and cross-platform Flutter client integrations. Demonstrated track record of slashing API latency by **42%**, orchestrating bidirectional real-time WebSocket pipelines for **50+ concurrent field technicians**, and designing Generative AI / RAG search systems using pgvector with sub-200ms latency. Dedicated adherent of Clean Architecture, domain-driven data models, idempotent API transactions, and automated test-driven development (**85%+ Pytest coverage**).

CORE TECHNICAL SKILLS TAXONOMY

Programming Languages:	Python 3.11+, Dart, SQL, Bash / Shell Scripting
Backend Frameworks & APIs:	FastAPI, Python AsyncIO, Uvicorn, Django, Flask, RESTful APIs, WebSockets, ASGI Channels, Pydantic v2, Swagger / OpenAPI
Databases & Data Storage:	PostgreSQL, MySQL, SQLite, Redis Caching, SQLAlchemy ORM, Alembic Migrations, B-Tree Indexing, Query Optimization, ACID
AI & Machine Learning:	Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), pgvector, LangChain, Vector Embeddings, Semantic Search
Cloud, DevOps & Tools:	Docker, Docker Compose, Linux / Ubuntu, AWS (EC2, S3), Nginx Reverse Proxy, Git, GitHub Actions (CI/CD), Postman
Architecture & Security:	Microservices, Clean Architecture, Event-Driven Architecture, High Concurrency, Non-blocking I/O, RBAC, JWT, OAuth 2.0
Testing & Code Quality:	Pytest, Unit Testing, Integration Testing, API Contract Testing, Test-Driven Development (TDD), Structured Error Envelopes
Frontend & Mobile:	Flutter, Dart, Cross-Platform Web & Mobile, Provider State Management, Offline Data Caching, WebSocket Client Sync

COMMERCIAL WORK EXPERIENCE

SPI Equipments

Oct 2023 – Present

Python Backend Developer & Full-Stack Engineer

Coimbatore, Tamil Nadu, India

Core backend and full-stack contributor engineering distributed enterprise maintenance automation platforms, real-time telemetry pipelines, and Flutter client applications.

- **42% Latency Reduction:** Refactored core RESTful API endpoints from synchronous blocking I/O to non-blocking FastAPI **AsyncIO** coroutines and async SQLAlchemy connection pooling, cutting average API latency by 42%.
- **Real-Time Event Streaming:** Architected a bidirectional WebSocket pub/sub notification pipeline delivering sub-second real-time service alerts to **50+ concurrent field technicians**, eliminating polling server overhead.
- **60% Query Execution Optimization:** Tuned complex relational schemas and query execution plans on PostgreSQL with composite B-tree indexing, cutting heavy join execution times by 60%.
- **35% Throughput Increase:** Modularized backend services using strict Pydantic v2 data models and centralized error envelopes, increasing request processing throughput by 35%.
- **85%+ Test Coverage:** Designed automated unit and integration test suites using **pytest** and **httpx.AsyncClient**, elevating code coverage to 85%+ with zero production regressions.
- **Cross-Platform Flutter Integration:** Developed and deployed cross-platform client mobile applications featuring offline-first local state caching and automated background synchronization.

KEY ENGINEERING PROJECTS

1. Service Management Application (Enterprise Field Dispatch)

`FastAPI • Python 3.11+ • WebSockets • PostgreSQL • SQLAlchemy • Flutter • Docker`

- Engineered full-lifecycle field maintenance platform connecting Admin, Service Managers, and Field Engineers for automated equipment service tracking.
- Bypassed polling latency with bidirectional WebSockets, delivering real-time ticket alerts and cutting technician response times by **45%**.
- Structured normalized PostgreSQL schema with composite B-tree indexing to process **10,000+** servicing history and part replacement logs.
- Implemented multi-tier Role-Based Access Control (RBAC) with JWT authentication separating field, management, and administrative privileges.

2. Restaurant Self-Ordering & Kitchen Display System (KDS)

`FastAPI • Python AsyncIO • WebSockets • MySQL • Flutter • Idempotency State Machine`

- Architected synchronized restaurant automation platform connecting table-side ordering kiosks with real-time Kitchen Display Systems.
- Eliminated duplicate order submissions during network dropouts using client-side idempotency keys and transactional database locks.
- Segmented WebSocket order broadcast channels by station (grill, fryer, cold prep), cutting kitchen routing confusion by **40%**.
- Delivered sub-second KOT propagation from customer tap to kitchen display with optimistic UI updates and auto-revert fallback.

3. Generative AI Medical & Industrial Knowledge Search Engine (RAG)

`Python 3.11+ • FastAPI • LangChain • OpenAI API • pgvector • PostgreSQL • Docker`

- Designed a Retrieval-Augmented Generation (RAG) platform performing contextual semantic search and Q&A across **10,000+** technical manual pages.
- Indexed high-dimensional vector embeddings in PostgreSQL via **pgvector**, achieving **<200ms** semantic similarity retrieval latency.
- Enforced strict prompt bounding eliminating model hallucinations and delivering verifiable citations to source engineering documentation.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering (B.E.) — Mechanical Engineering 2019 – 2023
KPR Institute of Engineering and Technology | Grade: 7.7 CGPA | Coimbatore, India

Python Full Stack Development Certification — Appin Technology (2023)
Verified Competencies: Python 3, Django, FastAPI, REST APIs, PostgreSQL, Docker, Git

CORE ENGINEERING PRINCIPLES & STANDARDS

Clean Architecture:	Strict separation of Domain Entities, Use Cases / Interactors, Repositories, and Delivery Gateways.
Concurrency & Scale:	Non-blocking asynchronous I/O (AsyncIO), event loops, connection pooling, and horizontal microservice scaling.
Resilience & Fail-Safes:	Idempotent request keys, automatic retry policies with exponential backoff, circuit breakers, and transactional rollback.
Security & Defense:	OAuth 2.0 / JWT bearer tokens, bcrypt credential hashing, role-based access control (RBAC), and SQL injection prevention via parameterized ORM queries.